

Archives of Agronomy and Soil Science

ISSN: 0365-0340 (Print) 1476-3567 (Online) Journal homepage: www.tandfonline.com/journals/gags20

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To cite this article: Mohammad Valipour (2014) Drainage, waterlogging, and salinity, Archives of Agronomy and Soil Science, 60:12, 1625-1640, DOI: 10.1080/03650340.2014.905676

To link to this article: https://doi.org/10.1080/03650340.2014.905676

Published online: 09 Apr 2014.

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Archives of Agronomy and Soil Science, 2014
Vol. 60, No. 12, 1625–1640, http://dx.doi.org/10.1080/03650340.2014.905676

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REVIEW ARTICLE

Drainage, waterlogging, and salinity

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(*Received 5 January 2014; accepted 12 March 2014*)

Nowadays, various areas in the world have faced waterlogging and salinity problems, which are intensified by a myriad of factors including use of wastewaters for irrigation, unsuitable cropping pattern, torrential rains and floods, lack of sufficient drainage, uncontrolled drainage, lack of adequate knowledge, wrong management decisions, very poor construction and rehabilitation rates of drainage systems, increase of irrigation systems without paying any attention to their adverse impacts on soil and quality of water resources, etc. One effective way to reduce waterlogging and salinity problems is drainage. It is necessary to do a comprehensive study on previous works to arouse awareness of drainage, waterlogging, and salinity and to avoid trial and error policies in these fields. The present paper aims to study drainage, waterlogging, and salinity in the different regions of the world and to present challenges and prospects for further research and making a better decision in this field. The results indicate that the frequency distribution of studies carried out in India, Australia, the USA, Pakistan, and other countries is 23%, 20%, 15%, 12%, and 30%, respectively. While frequency distribution is 45%, 30%, and 25% for focus on salinity, comprehensive studies, and focus on waterlogging, respectively, based on the previous investigations.

Keywords: drainage; improved drainage; rehabilitation; salinity; waterlogging

Relationship between drainage, salinity, and waterlogging

Compared to rainfed agriculture, irrigation is developing (Valipour 2013a, 2013b) and it increases waterlogging and salinity in agricultural fields. Therefore, the role of land drainage is vital and undeniable in the world. Figure 1 shows the status of drainage in recent decades based on Food and Agriculture Organization of the United Nations (FAO 2013).

Figure 1 shows the increase of drainage systems per cultivated area for Bangladesh, Jordan, Morocco, Pakistan, Tunisia, Lebanon, and Viet Nam. However, drainage systems have not been developed in Myanmar, Syria, Azerbaijan, and Kyrgyzstan, and even the percentage of drainage systems per cultivated area has been decreased for China, Armenia, Kazakhstan, Tajikistan, Turkmenistan, and Turkey. Additionally, Figure 2 shows the status of drainage in the previous years based on the International Drainage Database (IDD 2013).

As is shown in Figures 1 and 2, attention to drainage (values of more than 50%) is highlighted in Europe (Estonia, Italy, Slovenia, Denmark, Croatia, the UK, Ireland, Hungary, Netherlands, Lithuania, and Finland) in comparison with Asia (Korea, Turkmenistan, Uzbekistan, Pakistan, Japan, and Bangladesh), Americas (Suriname), and Africa (Egypt). The percent of total cultivated area drained is 12% and 17% based on the obtained data from FAO (2013) and IDD (2013), respectively. As use of drainage systems

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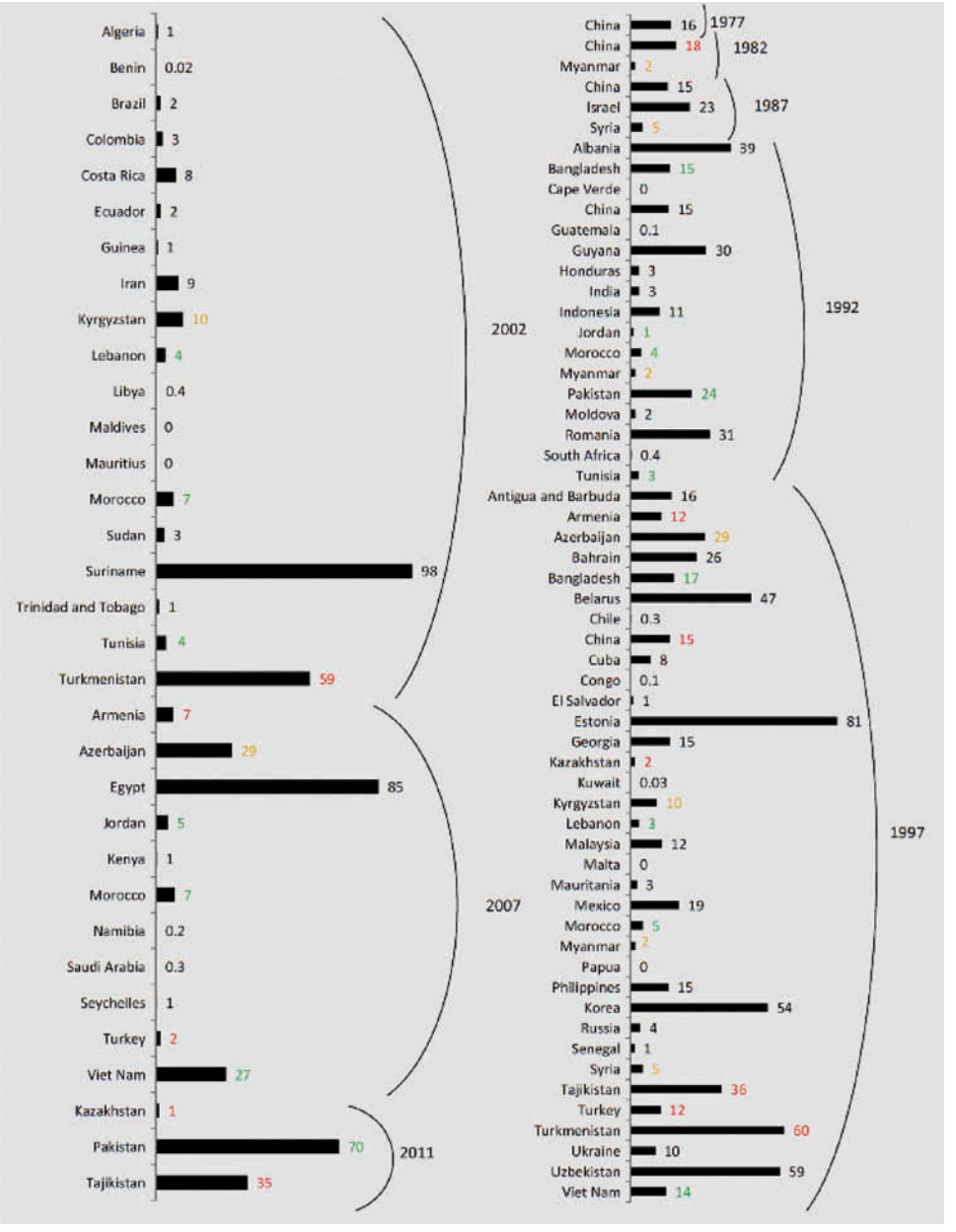


Figure 1. All available data (FAO 2013) of cultivated area drained (%).

is less than 20%, a considerable value of waterlogging and salinity could be seen in lands throughout the world (Figures 3 and 4).

Variations of salinity versus drainage

In Figure 3a, we can see the decrease of salinity in Egypt, Armenia, and Tajikistan; however, the value of salinity has not been decreased in Bangladesh, India, and

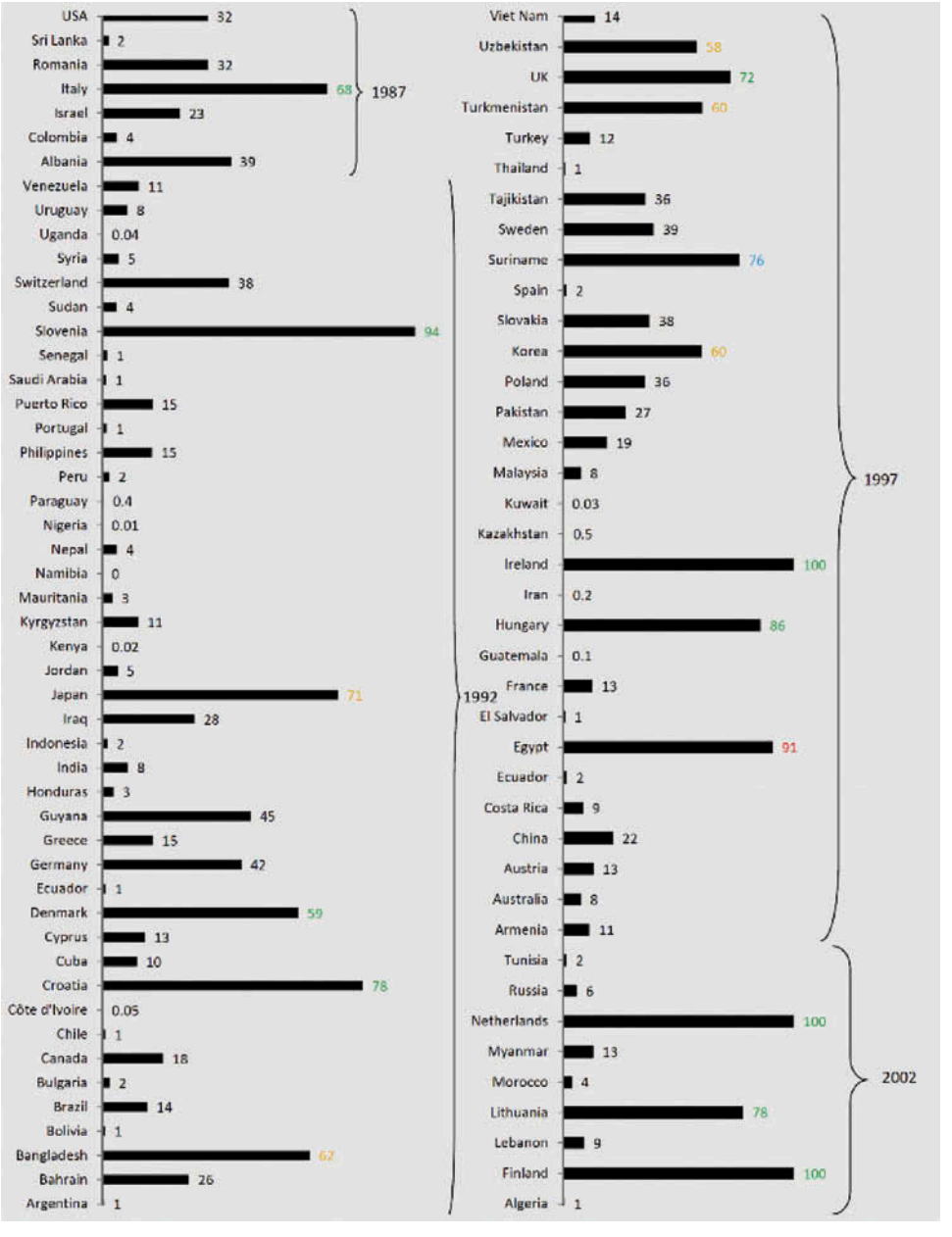


Figure 2. All available data (IDD 2013) of cultivated area drained (%).

Kyrgyzstan, and it has even increased in Azerbaijan, Kazakhstan, and Turkmenistan. So the management status of these countries is discussable. In Egypt, with a considerable decrease of salinity (more than 80%), we can see the highlighted role of drainage in recent years (7% salinity for 85% drainage in 2007 based on Figures 1 and 3a and 36% salinity for 91% drainage in 1997 based on Figures 2 and 4a). In Armenia, salinity and drainage have been decreased (both more than 40% based on Figures 1 and 3a) due to the proper management of drainage systems. Moreover, in Tajikistan, salinity and drainage have

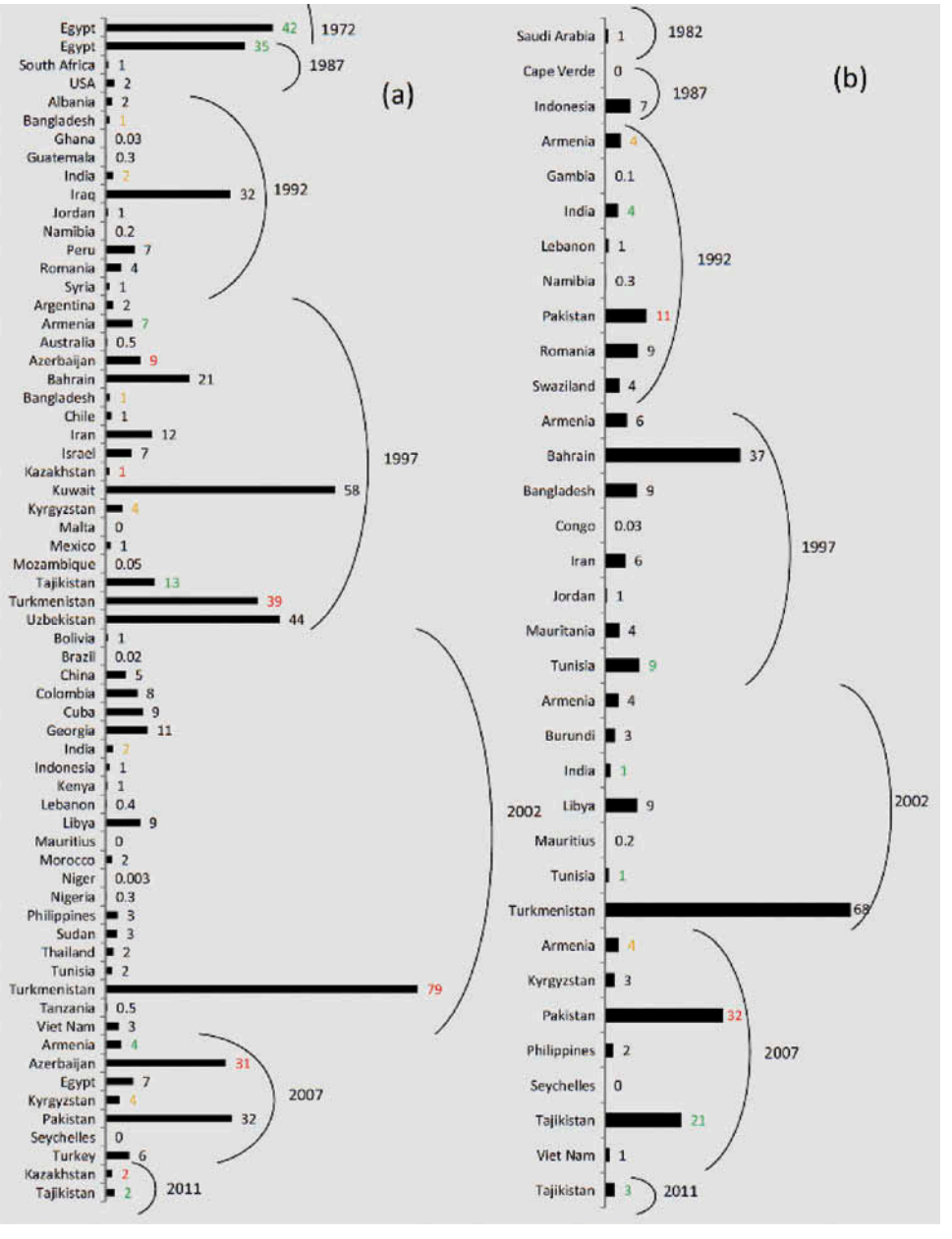


Figure 3. All available data (FAO 2013) of (a) cultivated area salinized (%) and (b) cultivated area waterlogged (%).

been decreased (more than 80% salinity and less than 3% drainage based on Figures 1 and 3a) due to adequate drainage (36% in 1997 based on Figures 1 and 2 as well as 35% in 2011 based on Figure 1) in recent years. In Bangladesh, we can see an increase in drainage (more than 10% based on Figure 1) during the 1990s and consequently lack of increase in salinity (Figure 3a). In Kyrgyzstan, we can assess the sustainability of drainage and consequently salinity (Figures 1 and 3a). In Azerbaijan, we can observe a drastic

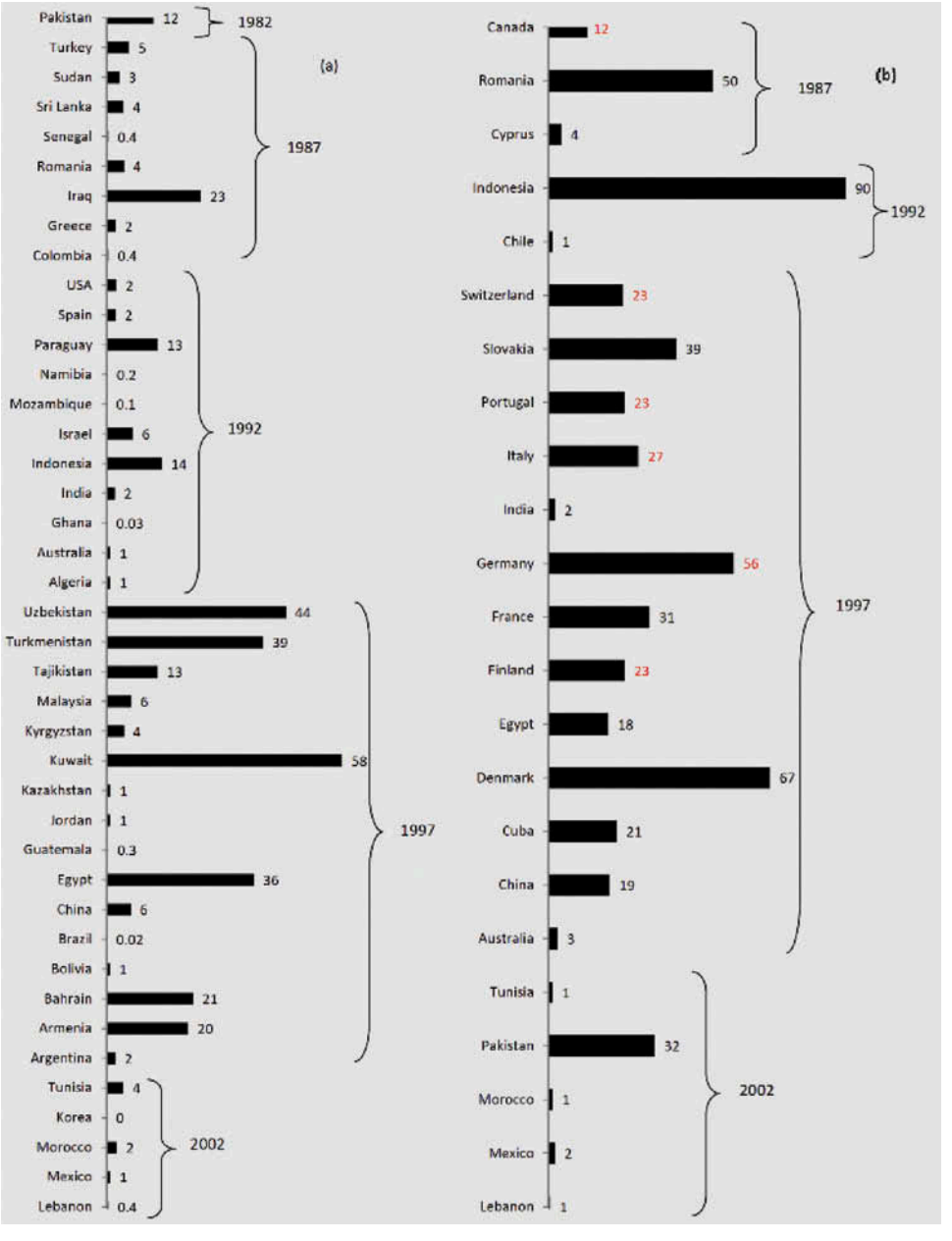


Figure 4. All available data (IDD 2013) of (a) cultivated area salinized (%) and (b) cultivated area waterlogged (%); as shown, there is not a considerable difference between IDD and FAO databases (Figures 3 and 4).

increase of salinity (more than 200% based on Figure 3a) due to the lack of attention to drainage systems and their development, according to Figure 1. In Kazakhstan, a decrease of drainage systems per cultivated area (Figure 1) and thereby an increase of salinity (Figure 3a) indicate management problems in the previous years. These conditions are also true in Turkmenistan with more than 100% increase of salinity (Figures 1 and 3a). In

1987, sufficient drainage and consequently low salinity are observable in the USA, Romania, and Colombia; however, the value of salinity is more than that of drainage in Sri Lanka. In 1992, sufficient drainage and consequently low salinity are observable in Albania, Jordan, Romania (like 1987), and Syria; however, the value of salinity is higher than that of drainage in Guatemala, Namibia, and Peru. In addition, insufficient drainage and consequently high salinity are observable in Iraq, Paraguay, and Indonesia. In 1997, sufficient drainage and consequently low salinity are found in Australia, Mexico, Malaysia, and China; however, the value of salinity is more than that of drainage in Chile and Guatemala (like 1992). In addition, insufficient drainage and consequently high salinity are noticeable in Iran, Bahrain, and Kuwait. In 2002, sufficient drainage and consequently low salinity are observable in Brazil, Lebanon, Morocco, and Seychelles. In addition, insufficient drainage and consequently high salinity are recognizable in Libya and Turkey.

Variations of waterlogging versus drainage

According to Figure 3b, we can see a decrease of waterlogging in India (without increasing salinity), Tajikistan (like its salinity condition), and Tunisia. However, the value of waterlogging has not been decreased in Armenia (although its salinity has been decreased) and it has even been increased in Pakistan. So the management status of these countries is debatable. In India, with a considerable decrease of waterlogging (more than 70%), we can see the highlighted role of drainage in recent years (Figures 1 and 2 as well as Figures 3b and 4b). In Tajikistan, waterlogging (like salinity) and drainage have been decreased (more than 80% waterlogging and less than 3% drainage based on Figures 1 and 3b) due to adequate drainage (36% in 1997 based on Figures 1 and 2 as well as 35% in 2011 based on Figure 1) in recent years. In Tunisia, drainage has been increased (Figure 1) and consequently waterlogging has been decreased (more than 80% according to Figure 3b). In Armenia, waterlogging (like salinity) and drainage have been decreased (6% salinity for 12% drainage in 1997 and 4% salinity for 7% drainage in 2007 based on Figures 1 and 3b) due to proper management of drainage systems. However, in Pakistan, although drainage has been increased (Figure 1), it is not adequate and the value of waterlogging has also been increased (Figure 3b). In 1987, insufficient drainage and consequently high waterlogging are observable in Romania (contrary to its salinity condition). In 1992, the value of waterlogging is more than that of drainage in Namibia (like its salinity condition). In addition, insufficient drainage and consequently high waterlogging are observable in Indonesia (like its salinity condition). In 1997, sufficient drainage and consequently low waterlogging are observable in Bangladesh (without increasing salinity), Congo, Egypt (according to the considerable difference between the values of drainage and waterlogging and like its salinity condition), and Australia (like its salinity condition); however, the value of waterlogging is more than that of drainage in Namibia (like 1992) and Mauritania. In addition, insufficient drainage and consequently high waterlogging are noticeable in Iran (like its salinity condition), Bahrain (like its salinity condition), Slovakia, France, Cuba, and China (contrary to its salinity condition). In 2002, sufficient drainage and consequently low waterlogging are found in Morocco and Lebanon (like their salinity condition); however, the value of waterlogging is more than that of drainage in Libya (with high value of salinity) and Mauritius. In addition, insufficient drainage and consequently high waterlogging are observable in Turkmenistan (like its salinity condition). In 2007, sufficient drainage and consequently low waterlogging are visible in Seychelles (like its salinity condition) and Viet Nam.

According to Figure 4b, although the value of waterlogging is considerable in Canada, Switzerland, Portugal, Italy, Germany, and Finland, they have not been discussed due to the lack of their values of drainage (Figures 1 and 2).

Drainage system rehabilitation and construction rates

The results show that management, development, rehabilitation, and monitoring of drainage systems are not acceptable in many countries according to their waterlogging and salinity conditions. Although available data are very limited for the study of rehabilitation and construction rates in the world (Figure 5), these data indicate a lack of global decision to improve drainage status and to solve waterlogging and salinity problems.

According to Figure 5, Europe is the only region with available data to study drainage management in comparison with the other regions (with the exception of Korea, Mexico, and Australia). It should be noted that in many countries drainage rehabilitation and/or construction rates are either very poor or the countries do not have any program for drainage rehabilitation and/or construction. Among all the countries in Figure 5a, Israel, Netherlands, Russia, Denmark, Korea, the UK, Sweden, Mexico, Lithuania, Austria, Poland, Australia, and Slovenia have sufficient drainage and consequently low salinity and/or waterlogging, according to Figures 1–4. However, Switzerland, Germany, Finland, and Italy, in spite of a considerable drainage, suffer from waterlogging and/or salinity problems (Figures 1–4). Moreover, in France and Portugal, we can see insufficient drainage and consequently high salinity and/or waterlogging (Figures 1–4). Therefore, drainage rehabilitation and/or construction rates need to increase in these countries. As rehabilitation and/or construction of drainage systems need(s) macro investments, in general, drainage systems are very expensive. Review of the results of previous researches and projects is necessary and inevitable to avoid trial and error policies and to reduce risk of failure for rehabilitation and/or construction projects in the future.

Review of previous researches

The impact of drainage on waterlogging and salinity has been studied in many researches. In this section, a review is done (based on a period of time from 1980s to 2013) on the most important obtained results according to their focuses on only waterlogging, only salinity, or a comprehensive study on both. In each part, only results of the more important researches are presented and similar works are not cited.

Focus on waterlogging

The significant impacts of mole drainage on soil properties indicate that mole drainage is very useful for reducing waterlogging in Thailand (Jha & Koga 1995). Using physical characteristics of the soil layers, it is only possible to simulate average waterlogging (it is not applicable for a point in the field) in Finland (Aura 1995). Yang et al. (2000) demonstrated that, to simulate water balance and drainage network system over a watershed, and to aid drainage management in an estuary floodplain environment, integrated drainage analysis system can be used effectively. In France, Bouarfa and Zimmer (2000) indicated that drain flow rates as well as kinetics of water table shape changes, as predicted by Boussinesq equation, are fully determined by combination of hydraulic and geometric properties of shallow systems. Singh and Shakya (2006) showed that gravity/cum/injection well (in comparison with the draining water through gravity flow) is a

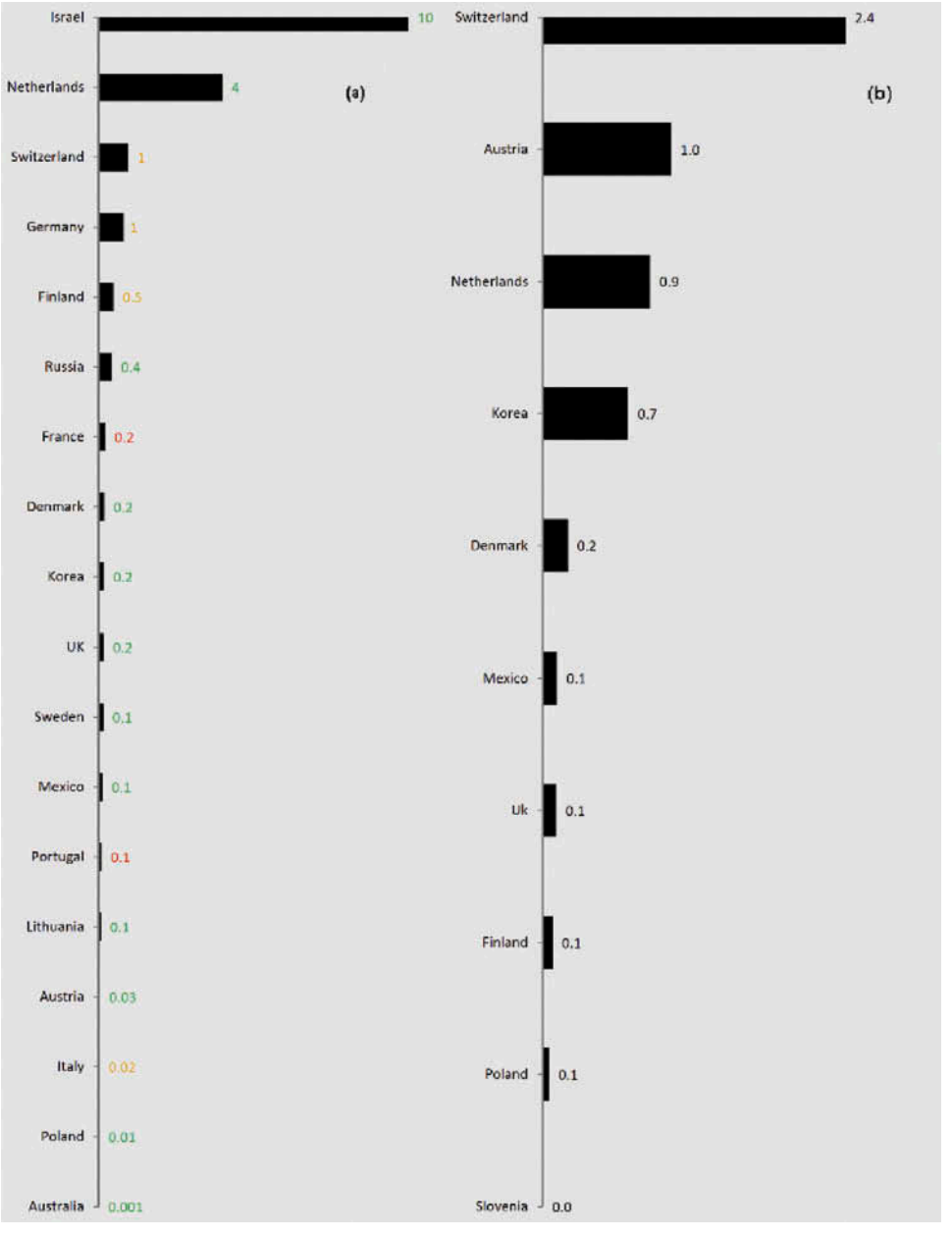


Figure 5. All available data (IDD 2013) for (a) annual drainage construction rate per cultivated area (%) and (b) annual rehabilitation rate per cultivated area drained (%).

solution for timely reclamation of waterlogging in India. The principal causes of drainage problems in Mali are saturation of drainage system with excess irrigation water and insufficient maintenance of drains (Vandersypen et al. 2007). According to different sizes of sites under serial biological concentration of salts (SBC) in Australia, the overall water table level remains at the depth of tile drain, which brings about a groundwater mound of 1.0 to 2.5 m on site and to a distance of 50 m laterally over time (Khan et al.

2007). In Pakistan, surface drains should be remodeled and maintenance should be carried out regularly for drainage of water from waterlogged areas (Bilal & Sarwar 2008). Chandio et al. (2013) indicated that managing waterlogging with the horizontal tile drainage systems (using a combined drainage system with tube wells plus horizontal drainage systems) is more beneficial in maintaining the water table within the desirable depths.

Focus on salinity

el-Din et al. (1987) evaluated the consequences of irrigation management policies on salt load leached to a drainage system using a finite element model. Improving drainage systems might be a cost-effective means of reducing eutrophication. Where phosphorus is a growth-limiting nutrient for crops, drainage system could not be recommended if load of dissolved orthophosphate phosphorus is critical for algae growth (Turtola & Paajanen 1995). Drainage systems in highly saline sodic soils cause 3–20% loss of total applied nitrogen through drainage system effluent in India (Singh, Bhattacharya, Nair, et al. 2002). Kahlown and Azam (2003) revealed that the maximum reduction in sodicity at upper depths was observed with manure and drainage management in Pakistan. Tanner et al. (2005) showed that annual mass removal of nitrogen from a drainage system in a surface flow wetland in New Zealand comprised 1% of the grazed pastoral catchment. Bakhsh et al. (2005) indicated that treatment impacts on NO_3–N losses with drainage water were found to be nonsignificant, because drain flows were found to be highly correlated with growing seasonal rainfall in the USA. The back propagation neural network (BPNN) modeling approach in India can be applied to drained fields to estimate salinity of drainage effluent and reclamation period with good accuracy using minimal input data. In addition, the SaltMod model (agro-hydro-salinity model for drainage, leaching of salts, and land rehabilitation) may be preferred over artificial neural network models for detailed salt balance component prediction in drained fields (Sarangi et al. 2006). The shallow drains had the lowest salt loads. Such salt loads could be reduced in Australia by managing deep pipe drains (Wahba & Christen 2006). According to Gormley and Campbell (2006), the estimation of solid transport above ground drainage system must follow code authorities under water conservation criteria. The increased surface runoff, due to shallow drainage, may raise concerns regarding increased concentrations of pollutants associated with runoffs such as ammonium–nitrogen, phosphorus, and pesticides in the USA (Singh et al. 2006). The Water Flow and Balance Simulation Model (WaSiM) model is a useful and user-friendly tool to assess the hydraulic performance of drainage system or to design drainage systems for semi-arid monsoon climates in India (Hirekhan et al. 2007). Singh et al. (2007) revealed that leaching of drainage system helped in improving the fertility status of saline soils by promoting ammonium storage on clay complexes in India. Opportunistic summer forage cropping, as drainage management, helped to prevent further dryland salinity in Southeast Australia (Wang et al. 2008). Under current irrigation systems, controlled drainage reduced the discharge by 50% to 60%; salinity from controlled fields was higher than that from conventional drainage systems in China (Luo et al. 2008). Welderufael and Woyessa (2009) indicated that the soils at the root zone were not sufficiently remedied from salinity by improved drainage in the Central Highlands of Ethiopia. Luo et al. (2010) claimed that controlled drainage had the greatest potential to reduce drainage volumes as well as nitrogen losses in the USA. Ale et al. (2012) showed that drain spacing and soil parent materials had more impact on the reduction in nitrate loads in drainage water than cropping pattern in the USA. All drainage salinity management strategies are likely to further degrade the Blackwood River in Australia

because of higher salinity levels (Ali et al. 2012). Shao et al. (2012) indicated that drainage systems were significant in reducing soil salinity and suppressing resalinization for secondary salinization greenhouse soil in China. The subsurface drainage system has proven to be effective in improvement of soil properties in Italy (Mastrocicco et al. 2013).

Comprehensive studies (assessment of both waterlogging and salinity)

The model developed by Alaouze and Fitzpatrick (1989) provided a framework for collection of data on river protection, land waterlogging, and salinity schemes. In the UK, the analysis endogenously accounted for farmers adjusting their activity mixes over time in order to continue maximizing profits in face of changing levels of soil salinity and waterlogging. Assuming that farmers do not attempt to minimize their net farm results influences the future financial status of salinity and waterlogging, and it is highly expected that drainage plays critical roles (Jones & Marshall 1992). Kapoor (2000) showed that biodrainage can be an option to control waterlogging and soil salinization. Regarding irrigation water, which is of a good quality, total minerals removed annually by crop and forest biomass can match the total annual import of minerals with irrigation water. The results indicate that forest plantations in no more than in about 10% of the area, can provide satisfactory insurance against waterlogging in India. Presenting a sustainable solution for disposal of drainage effluent, financial feasibility of drainage in saline and waterlogged lands, which looks favorable, depends on the fact that sufficient water is available for irrigation and leaching in India (Datta et al. 2002). Singh, Bhattacharya, Singh, et al. (2002) showed that, using SaltMod, the salinity of root zone could be predicted more accurately than the quality of drainage water and water table depths in India.

The groundwater table, which used to be almost at the ground surface during October to February before installation of drainage systems, could be lowered by 0.20 to 0.35 m due to the installation of drainage systems. Also, a total of 166 tons of salts have been disposed through drainage system during the period of 3 years in India (Srinivasulu et al. 2005). The existing design criteria and procedures result in the discharge of excessive drainage water volumes and salt loads with a negative effect on the quality of receiving surface or groundwater. Studies in the USA indicated that water-table control was effective on a 60 hectare field. In addition, field studies in Australia showed that water-table control was effective in vineyards without negative effects (Ayars et al. 2006). The combination of shallow and deep drains meets the requirements for waterlogging control and leaching of the shallow layer using a shallow drain and providing water-table control to prevent resalinization using deep drain in Australia (Hornbuckle et al. 2007). The left and right bank outfall drains can play a major role in reclaiming waterlogged and saline soils and providing an outfall for evacuating the salt from existing and future drainage projects in Pakistan (Qureshi et al. 2008). Woody tree species like eucalyptus were recommended in waterlogged sodic soil, where water table is at the surface in India (Shakya & Singh 2010).

General discussion

In the following, a list of major problems and perspectives is presented.

Problems

Much of the apparent chaos in spatial patterns of soil salinity is related to the interacting drainage, irrigation, and tillage management practices (Rhoades et al. 1997).

The traditional approach to drain design is to install drains as deep as possible, which allows wider lateral spacings to be used. This design approach has been found to increase the amount of salt and toxic chemicals in drainage water compared with shallow-depth installations and smaller spacings. In some areas, deeper shallow groundwater contains higher concentrations of salt, and in some cases, selenium, arsenic, and boron compared to concentrations in shallow groundwater near the water table. Using deep drain depths and wide spacings, deeper groundwater was found to contribute more significantly to the total drain flow compared with the shallow groundwater. This is because the wide spacings and deep drains caused a ground water flow pattern that extended deeper into the poorer-quality groundwater and resulted in water being displaced toward the drain (Hanson et al. 1999).

A measuring season with both too few observations and too small range of water tables and discharges gives less reliable information (Rimidis & Dierickx 2003).

The use of poor-quality drainage water for irrigation without suitable amendments should be discouraged as it can increase soil salinity (Kahlown & Azam 2003).

Inadequate drainage in irrigation schemes is a growing concern due to consequences on agricultural production and environment (Vandersypen et al. 2007).

Subsurface drainage can only be successful in controlling salinity if sufficient good-quality irrigation water or monsoon rainfalls are available for leaching (Ritzema et al. 2008).

Subsurface drainage induces a decrease in the intensity of erosion and an increase in the intensity of lixiviation and eluviation, as well as specific redox processes, compared to similar undrained soils, and Fe, Mn, and particle translocations and losses from the soil are of particular concern; these processes do not affect the whole soil to a similar extent due to gradients of water fluxes and waterlogged conditions existing from the immediate vicinity of the drain line to the mid-drain position, and heterogeneities can then be expected at the 10 meter scale in the evolution of the horizon above the hydraulic barrier; and soil reacts over time scales of a few decades to the perturbations induced by subsurface drainage (Montagne et al. 2009).

Perspectives

Improved drainage is also urgently required to arrest land degradation and production loss caused by waterlogging and salinity in irrigated areas of arid zone (Smedema & Ochs 1998).

Improvements in the definition of property rights to scarce resources and more appropriate pricing of irrigation and drainage inputs may reduce the rate of increase in saline and waterlogged lands. Many of these improvements can be implemented while installing or renovating water delivery and subsurface drainage systems, and when transferring responsibility for operation and maintenance from central government agencies to local irrigation districts (Wichelns 1999).

The water must be applied with a high degree of uniformity to prevent excessive deep percolation in some parts of the field, which could cause waterlogging. Irrigation systems appropriate to this management approach include drip irrigation, linear move sprinkler systems, center-pivot sprinkler systems, and low energy precise application (LEPA) irrigation machines (Hanson et al. 1999).

The water quality of drains installed at relatively shallow depths with a smaller spacing may contain less salt, thus reducing salt loads of the drainage water. At the same time, the volume of discharged drainage water may be less compared with deeper, wider-spaced drains. This is because much of the flow pattern does not extend as deep into the poorer-quality groundwater. As a result, the relatively better-quality shallow groundwater was skimmed off near the water table and contributed more to the total discharge of subsurface drainage (Hanson et al. 1999).

Improved drainage is also urgently required to arrest land degradation and production loss caused by waterlogging and salinity in arid zone irrigated lands (Smedema & Ochs 1998).

The proper time for installing a deep drainage system depends on how rapidly the water table rises after the start of irrigation (Datta et al. 2000).

It is emphasized that improved drainage can contribute to establishing a more diversified, competitive and sustainable type of agriculture, enhance sanitary and public health conditions and generally contribute to rural development, rural well-being, and poverty alleviation (Smedema et al. 2000).

The assessment of economic viability of investment on land drainage was purely based on the actual land-use pattern, and crop yield monitored after the drainage installation is still increasing. The optimum and realistic effects on the increase in land use, crop yields, and consequently on the net returns as a result of reduction in soil salinity and the water-table depth are yet to be seen. It would take some more time to achieve optimum yields. It is believed that the indicators of economic performance of the investment on drainage would further improve (Indo-Dutch Network Project 2002).

To make the shift from the manual to mechanized installation of the drain, a number of problems had to be solved (Ritzema et al. 2006).

The efforts to improve drainage practices will facilitate the further introduction of pipe drainage in the world and this contributes to a better, more sustainable use of the world's precious land and water resources (Ritzema et al. 2006).

It is imperative that fields having subsurface drainage must continue to operate the system and keep the reclaimed land under cultivation to maintain soil health on a long-term basis (Singh et al. 2007).

It is recommended to prepare drainage policy papers at both national and state levels emphasizing the time-bound action plans to reclaim saline and waterlogged fields (Ritzema et al. 2008).

Denitrification bioreactors are a promising nitrate mitigation option for drainage systems in a variety of agricultural locations (Christianson et al. 2011).

Controlled drainage with low-grade weirs is a strategy for surface drainage ditches that will increase hydraulic retention capacity, potentially improve conditions for biogeochemical transformations, and thus enhance nitrate concentration and load reductions over drained systems (Kroger et al. 2012).

Conclusion

According to the previous studies, the impact of drainage on waterlogging and salinity is considerable from various aspects and we cannot adopt optimum decisions without a complete review. Figure 6 shows a brief study of drainage, waterlogging, and salinity in the different regions.

Compared to waterlogging, the rate of salinity problems is higher and they are required to receive more attention (Figure 6) in the world. In India, although cultivated

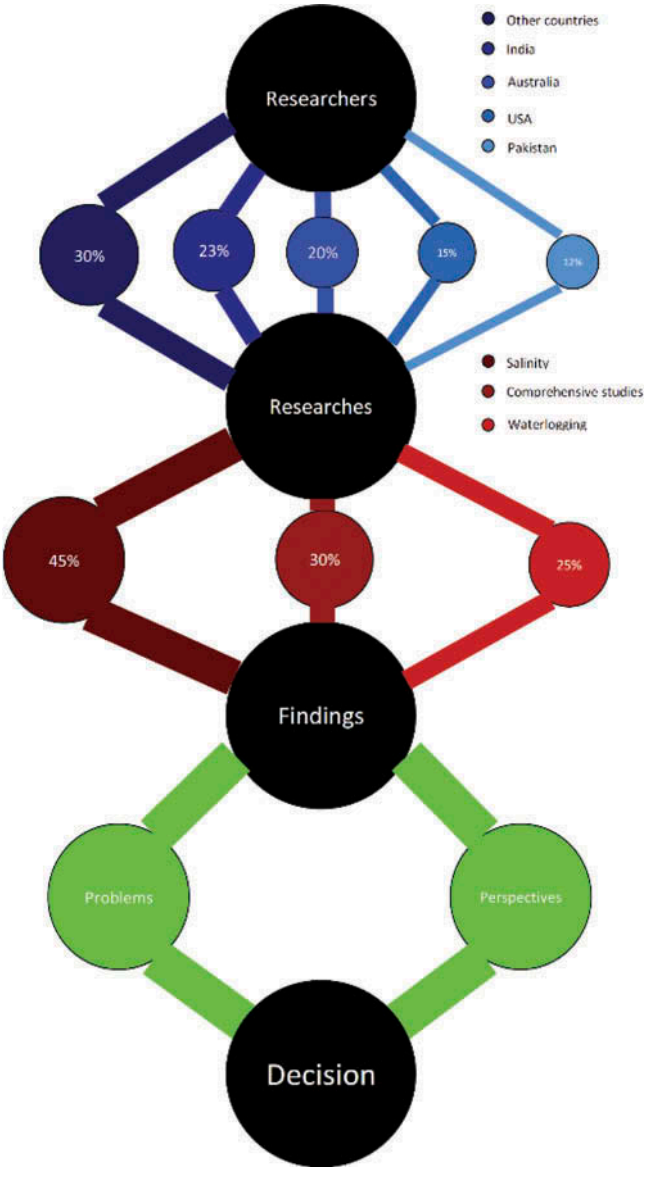


Figure 6. Study of drainage, waterlogging, and salinity in the different regions.

area drained is less than 10% (Figures 1 and 2), several studies have been done (23%) and we can see that the value of salinity has not been increased, but the value of waterlogging has been decreased (more than 70%) (Figures 3a and 3b, respectively). This shows clearly the importance of research on the relationship among drainage, waterlogging, and salinity. In Australia with 20% of researches, and although cultivated area drained is less than 10% (Figure 2), we can see sufficient drainage and consequently low salinity and waterlogging according to Figures 3 and 4 as well as a few percent of construction rate according to Figure 5. This emphasizes again the importance of research on the relationship among drainage, waterlogging, and salinity. In the USA with 15% of researches, we can find 32%

drainage (Figure 2) and low salinity (Figures 3 and 4) for it. Finally, in Pakistan with 12% of researches, and although a highlighted increase has occurred (almost 200% according to Figure 1), the value of waterlogging has also increased (Figure 3a). This underlines the need for further studies and also attention to rehabilitation of drainage systems in Pakistan. Although the focus on only salinity or only waterlogging is useful, an increase of comprehensive researches (assessment of both) in comparison with separate investigations can help to achieve more reliable and extensive findings and finally make a better decision for reduction of waterlogging and salinity using an improved and controlled drainage system. The mentioned cases results emphasized the need of proper recognition of problems and perspectives of drainage, waterlogging, and salinity to design a bright future for more attention to these issues.

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